MODELING INFLATION AND DEFLATION

Learning Outcome

- explain how to forecast industry and company sales and costs when they are subject to price inflation or deflation

Inflation and deflation (i.e., general increase and decrease in the prices of goods and services) can significantly affect the accuracy of forecasts for a company's future revenue, profit, and cash flow. The impact of inflation or deflation on revenue and expenses differs from company to company. Even within a single company, the impact of inflation or deflation is generally different for revenue and expenses categories.

Some companies are better able to pass on higher input costs by raising the prices at which they sell their output. The ability to pass on price increases can be the result of, for example, strong branding (Coca-Cola) or proprietary technology (Apple). Companies that are well positioned to pass on price increases are, in turn, more likely to have higher and more stable profits and cash flow, relative to competitors.

We first consider the impact of inflation on sales and then on costs.

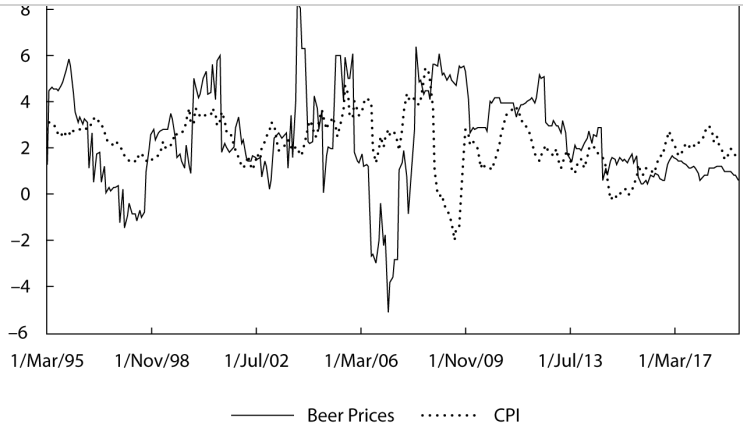
Sales Projections with Inflation and Deflation

The following analysis addresses the projection of industry sales and company sales in the presence of inflation.

Industry Sales and Inflation or Deflation

Most increases in the cost of inputs, such as commodities or labor, will eventually result in higher prices for end products. Industry structure can be an important factor in determining the relationship between increases in input costs and increases in the price of end products. For example, in the United States, the beer market is an oligopoly, with one player, AB InBev, controlling almost half of the market. Moreover, the three-tier structure of the US beer market, in which the producers (the brewers) must use a third party (the wholesalers) to get their products (beer) to the consumers (bars, restaurants, and retailers) results in a fragmented customer base because brewers are not allowed to deliver directly to the end consumer but rather must use wholesale distributors. These wholesalers often differ state by state. Large nationwide retailers, such as Walmart, still must negotiate with several different wholesalers instead of using their dominant national market position to negotiate directly with the brewers. The industry structure in the United States has likely contributed to increases in beer prices roughly in line with the US Consumer Price Index. In other words, beer prices have generally risen during years of inflation in input costs and decreased when costs have eased (though there have been brief exceptional periods where the opposite has occurred). If necessary, US brewers have been able to increase prices to compensate for costs of inflation. In contrast, European beer companies distribute through a more concentrated customer base—namely, such dominant retail outlets as Carrefour, Tesco, and Ahold—which results in a weaker pricing position for the brewers. Also, the European market lacks an overall dominant brewer. As a result of the industry structure and the lack of underlying volume growth, changes in beer prices in Europe have been on average 100 bps less than customer inflation.

Exhibit 25: US General Inflation and Inflation in Beer Prices



▶ Expand / collapse Extended Description

Source: US Bureau of Labor Statistics.

A company's efforts to pass on inflation through higher prices can have a negative impact on volume if the demand is price elastic, which is the case if cheaper substitutes are available. If selling prices could be increased 10 percent while maintaining unit sales volume to offset an increase of 10 percent in input costs, gross profit margin percentage would be the same but the absolute amount of gross profit would increase. In the short term, however, volumes will usually decline as result of a price increase. The decline would depend not only on the price elasticity of demand but also on the reaction of competitors and the availability of substitutes. Lower input costs also make lower consumer prices possible. The first competitor to lower prices will usually benefit with an uptick in volume. Competitors react quickly, however, resulting in a short-term benefit. The price–volume trade-off can make accurate revenue projections difficult. In an inflationary environment, raising prices too late will result in a profit margin squeeze but acting too soon could result in volume losses. In a deflationary environment, lowering prices too soon will result in a lower gross margin, but waiting too long will result in volume losses.

In the highly competitive consumer goods market, pricing is strongly influenced by movements in input prices, which can account for half of the COGS. In some time periods, customers' price sensitivity has resulted in a strong inverse relationship between volume and pricing. For example, Exhibit 26 illustrates Unilever's annual underlying volume and price growth from 2001 to 2020. Increased input prices for packaging, wheat, and milk forced Anglo-Dutch consumer staple company Unilever to increase prices for its products significantly in 2008. Consequently, volumes deteriorated. But as raw material prices fell in 2009–2010, the company's prices were lowered and volumes recovered strongly. As the company started to increase prices in 2011, volume growth once again slowed. In 2016, the company faced challenging conditions in several emerging markets as currency-devaluation-led cost increases led to weaker volumes. Both volume and price growth have moderated to low-single digit growth rates, also exhibiting lower volatility.

Exhibit 26: Unilever Overall Revenue Growth by Percentage Change in Volume and Price

▶ Expand / collapse Extended Description

Sources: Unilever PLC filings.

Company Sales and Inflation or Deflation

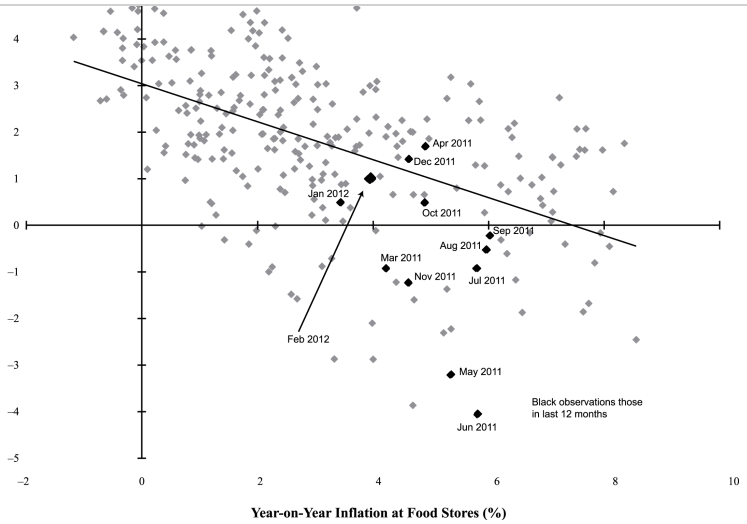
Revenue projections in a model are based on the expected volume and price development. Forecasting revenue for a company faced with inflation in input costs requires some understanding of the price elasticity of the products, the different rates of cost inflation in the countries where the company is active, and, if possible, the likely inflation in costs relevant to a company's individual product categories. Pricing strategy and market position are also important.

The impact of higher prices on volume depends on the price elasticity of demand (i.e., how the quantity demanded varies with price). If demand is relatively price inelastic, revenues will benefit from inflation. If demand is relatively price elastic (i.e., elasticity is greater than unit price elasticity), revenue can decline even if unit prices are raised. For example, a regression of volume on food inflation in UK food stores from 1989 to 2012 (shown in Exhibit 27) gives a regression slope coefficient of −0.398. (For every increase by 1 percentage point in year-on-year food prices, year-on-year sales decreased by approximately 0.4 percent.)

An analyst covering UK food retailers can use this information when building forecast profit models. By assuming an expected level of food inflation, volume growth can be estimated and revenue calculated.

Exhibit 27:

UK Relationship between Food Inflation and Volume, January 1989–February 2012



▶ Expand / collapse Extended Description

Source: Based on data from Datastream. Analysis is the authors'.

The expected pricing component for an international company should consider the geographic mix of its revenues to reflect different rates of inflation among countries. Of course, strategy and competitive factors, in addition to inflation in input costs, play roles in price setting.

AB InBev's volume growth and pricing have been more robust in emerging markets, for example, thanks to strong demand for its new beer products. The impact of inflation is also an important factor. In its Latin America South division, which then mainly consisted of Argentina, the brewer reported strong 24.7 percent organic revenue growth in 2011, of which only 2.1 percent was driven by volume and the remainder by price. As costs increased in line with revenues, operating margin remained more or less stable, and organic operating profit growth was high at 27 percent. With only a limited negative currency impact, reported operating profit increased 24 percent in US dollars.

High inflation in a company's export market relative to a company's domestic inflation rate generally implies that the export country's currency will come under pressure and any pricing gain could be wiped out by the currency losses. The strong pricing increases AB InBev reported in its Latin America South division were clearly driven by input price inflation. The absence of a negative currency impact should be seen as a positive surprise but not as a typical outcome. A country's currency will usually come under pressure and depreciate if high rates of inflation persist for an extended period.

Most analysts adjust for recent high inflation in foreign countries by assuming a normalized growth rate for both revenues and costs after one or two years. This constant currency growth rate is based on an underlying growth rate assumption for the business. This approach can understate revenues in the short term. Other analysts reflect in their forecasts the high impact of inflation on revenues and expenses and adjust growth rates for the expected currency (interest rate parity) impact. This approach is also imperfect given the difficulty in projecting currency rates.

Identifying a company's major input costs provides an indication of likely pricing. For a specialist retail bakery chain, for example, the impact of increased grain prices will be more significant than for a diversified standard supermarket chain. Consequently, it seems logical that the bakery is likely to increase its prices by a higher percentage than the grocer in response to increased grain prices.

Company strategy is also an important factor. Faced with rising input prices, a company might decide to preserve its margins by passing on the costs to its customers, or it might decide to accept some margin reduction to increase its market share. In other words, the company could try to gain market share by not fully increasing prices to reflect increased

affected customers (e.g., restaurants, private clubs, schools, nursing homes). On the other hand, in 2011 and 2012, the large French cognac houses substantially increased the prices of their products in China to reduce strong demand. Because older cognac generates a higher price, it can be more profitable to build an inventory of vintage cognac rather than maximizing short-term volumes.

EXAMPLE 8

Passing on Input Cost Increases or Not

Category
Introduction to Financial Statement Modeling

Four food retail analysts are assessing the impact of a potential increase in input costs on the global supermarket chain Carrefour. In this hypothetical scenario, they believe that rising oil prices and packaging prices will affect many of the company's suppliers. They believe that Carrefour is likely to be confronted with 4 percent inflation in its COGS (with stable volume). The analysts have their own expectations about how the company will react. Exhibit 28 shows Carrefour's 2020 results, and Exhibit 29 shows the four analysts' estimates of input prices, volume growth, and pricing for the following year. Both exhibits are in the Example 8 worksheet in the downloadable Microsoft Excel workbook.

Exhibit 28: Carrefour Data (euro millions, unless noted)

	2020
Total revenue	72,150
COGS	56,705
Gross profit	15,445
Gross margin	21.4%

Exhibit 29: Four Analysts' Estimates of Carrefour's Reaction to Inflation

	A	B	C	D
Price increase for revenues	0.00%	2.00%	3.00%	4.00%
Volume growth	5.00%	2.00%	1.00%	−4.00%
Total revenue growth	5.00%	4.04%	4.03%	−0.16%
Input costs increase	4.00%	4.00%	4.00%	4.00%

1. What are each analyst's predictions for gross profit and gross margin?

Hide Solution

Solution:

The results for each analyst are shown in Exhibit 30 and the Example 8 worksheet in the downloadable Microsoft Excel workbook. For Analyst B, revenues increase 4 percent [= (1.02 × 1.02) - 1] and COGS 6.1 percent [= (1.02 × 1.04) - 1]. The difference between the calculated revenue and COGS is the new gross profit and gross margin is gross profit as a percentage of revenue.

Exhibit 30: Results for Analysts' Predictions (EUR millions, unless noted)

Total revenue	72,150	75,758	5.0%	75,065	4%	75,058	4.0%	72,035
COGS	56,705	61,922	9.2%	60,153	6%	59,563	5.0%	56,614
Gross profit	15,445	13,836	–10%	14,912	–3%	15,495	0%	15,420
Gross margin	21.4%	18.3%		19.9%		20.6%		21.4%

2. Which analyst has the highest forecast for gross margin?

Hide Solution

Solution:

The highest gross margin is projected by Analyst D, who assumes that selling prices would increase by 4 percent to offset rising input costs and keep gross margin stable from the 2020 level.

3. Which analyst has the highest forecast for gross profit?

Hide Solution

Solution:

The highest gross profit is projected by Analyst D.

Cost Projections with Inflation and Deflation

The following analysis addresses the forecasting of industry and company costs in the presence of inflation and deflation.

Industry Costs and Inflation or Deflation

Familiarity with the specific purchasing characteristics of an industry can also be useful in forecasting costs. For example, long-term price-fixed forward contracts and hedges can delay the impact of price increases. Thus, an analyst forecasting costs for an industry in which companies customarily use such purchasing practices would incorporate any expected input price fluctuations more slowly than they would for an industry in which the participants do not use long-term contracts or hedges.

Monitoring the underlying drivers of input prices can also be useful in forecasting costs. For example, weather conditions can have a dramatic impact on the price of agricultural products and consequently on the cost base of industries that rely on them. An analyst observing a particular weather pattern might thus be able to incorporate this information into forecasts of costs.

How inflation or deflation affects an industry's cost structure depends on its competitive environment. For example, if the participants within the industry have access to alternative inputs or are vertically integrated, the impact of volatility in input costs can be mitigated. Jacobs Douwe Egberts (JDE) is a coffee company that has been facing high and volatile coffee prices. However, its coffee is a blend of different kinds of beans. By shifting the mix slightly, JDE can keep both taste and costs constant by reducing the amount of the more expensive types of coffee beans in the blend. But if all supplier countries significantly increase the price of coffee simultaneously, JDE cannot use blending as an offset and will be confronted with overall higher input costs. To sustain its profitability, JDE will have to increase its prices to its clients. But if competition from other companies, such as Nestlé (Nespresso, Dolce Gusto, Nescafe) makes it difficult to increase prices, JDE will have to look for alternatives if it wants to keep its profit margins stable. An easy solution for the short term could be reducing advertising and promotional (A&P) spending, which usually

grain harvest, causing prices for malting barley, a major input for beer, to increase significantly. Carlsberg, as the largest Russian brewer at that time, was particularly hard hit because it had to pay more for its Russian barley and also needed to import grain into the country, incurring additional transportation costs. By increasing imports from Western Europe, Carlsberg also pushed up barley prices in this region, affecting the cost base of other Western European brewers.

Company Costs and Inflation or Deflation

In forecasting a company's costs, it is often helpful to segment the cost structure by category and geography. For each item of cost, an assessment should be made about the impact of potential inflation and deflation on input prices. This assessment should take into account the company's ability to substitute cheaper alternatives for expensive inputs or to increase efficiency to offset the impact of increases in input prices. For example, although a jump in raw material prices in 2011 caused Unilever's and Nestlé's gross margins to fall sharply (by 110−170 bps), increases in operational efficiencies, such as reducing advertising spending, enabled both companies to achieve slightly higher overall operating profit margins that year. Example 9 shows the use of common size (percent-of-sales) analysis of inflation in input costs.

EXAMPLE 9

Inflation in Input Costs

Two fictional consumer staple companies—chocolate and sweets specialist "Choco A" and a food producer "Sweet B"—have costs that are constantly affected by inflation and deflation. Exhibit 31 (see the Example 9 worksheet in the downloadable Microsoft Excel workbook) presents a common size analysis.

Exhibit 31: Common Size Analysis for Sweet B and Choco A

	Sweet B	Choco A
Net sales	100%	100%
COGS	50%	36%
Gross margin	50%	64%
SG&A	31%	47%
Depreciation	3%	4%
EBIT	16%	13%
Raw materials	22%	22%
Packaging	12%	10%
Other COGS	16%	4%
Total COGS	50%	36%

Assume inflation of 10 percent for all costs (except depreciation) and that the companies are not able to pass on this increase through higher prices (total revenues will remain constant).

1. Calculate the gross profit margin for each company. Which company will experience the greater reduction in gross profit margin?

Hide Solution

Solution:

percent, as shown in Exhibit 31. After the 10 percent increase in COGS to 1.10 × 50% = 55%, Sweet B's gross margin will fall to 45 percent, as shown in Exhibit 32. Sweet B's resulting gross margin of 45 percent represents a proportional decline of 10 percent from the initial value of 50 percent. In contrast, the proportional decline in Choco A's gross margin is approximately 4%/64% = 6%.

Exhibit 32: Effect of Cost Inflation

	All Costs (Except Depreciation) + 10%		Raw Materials + 10%	
	Sweet B	Choco A	Sweet B	Choco A
Net sales	100%	100%	100%	100%
COGS	55%	40%	52%	38%
Gross margin	45%	60%	48%	62%
SG&A	34%	52%	31%	47%
Depreciation	3%	4%	3%	4%
EBIT	8%	5%	14%	11%

2. Calculate the operating profit margin for each company. Which company will experience the greater reduction in operating profit (EBIT) margin?

Hide Solution

Solution:

Choco A has higher overall costs than Sweet B, primarily as a consequence of its high SG&A expenses. Choco A's operating profit margin will drop to approximately 5 percent, as shown in Exhibit 32, representing a proportional decline of approximately 62 percent compared with a proportional decline of approximately 8%/16% = 50% for Sweet B.

3. Assume inflation of 10 percent only for the raw material costs (reflected in COGS) and that the companies are not able to pass on this increase through higher prices. Which company will be more affected negatively in terms of gross profit margin and operating profit margin?

Hide Solution

Solution:

The company with the higher raw material expense component will experience the more negative effect. In this case, raw materials represent 22 percent of net sales for both Sweet B and Choco A. Gross margin and operating margin will decline by 220 bps for both. This impact is more severe on gross margin on a relative basis for Sweet B (2.2%/50% = 4.4% decline) than for Choco A (2.2%/64% = 3.4% decline). But the relative effect on operating margin will be more severe for Choco A (2.2%/13% = 16.9% decline) than for Sweet B (2.2%/16% = 13.8%).

Discuss
Filter
Discussion
What do you want to discuss... (required)

© 2025 CFA Institute. All Rights Reserved.

Category

Introduction to Financial Statement Modeling

Rate Your Confidence

Category

Introduction to Financial Statement Modeling

Rate Your Confidence

Category

Introduction to Financial Statement Modeling

Rate Your Confidence

Category

Introduction to Financial Statement Modeling

Rate Your Confidence

Category

Introduction to Financial Statement Modeling

Rate Your Confidence

Category

Introduction to Financial Statement Modeling

Rate Your Confidence

Category

Introduction to Financial Statement Modeling

Rate Your Confidence

Category

Introduction to Financial Statement Modeling

Rate Your Confidence

Category

Introduction to Financial Statement Modeling